

WEEK 6 REPORT

Healthcare Dashboard: DAX Measures & Visual Layout

(Power BI Desktop)

Submitted by
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1. Introduction

This week I used the clean healthcare data model from Week 5 (Dim_Patient, Dim_Admission, Fact_Billing) to build an actual dashboard. I wrote the DAX measures for the KPIs first, then arranged cards, charts and slicers on the report page so that the hospital data can be explored visually.

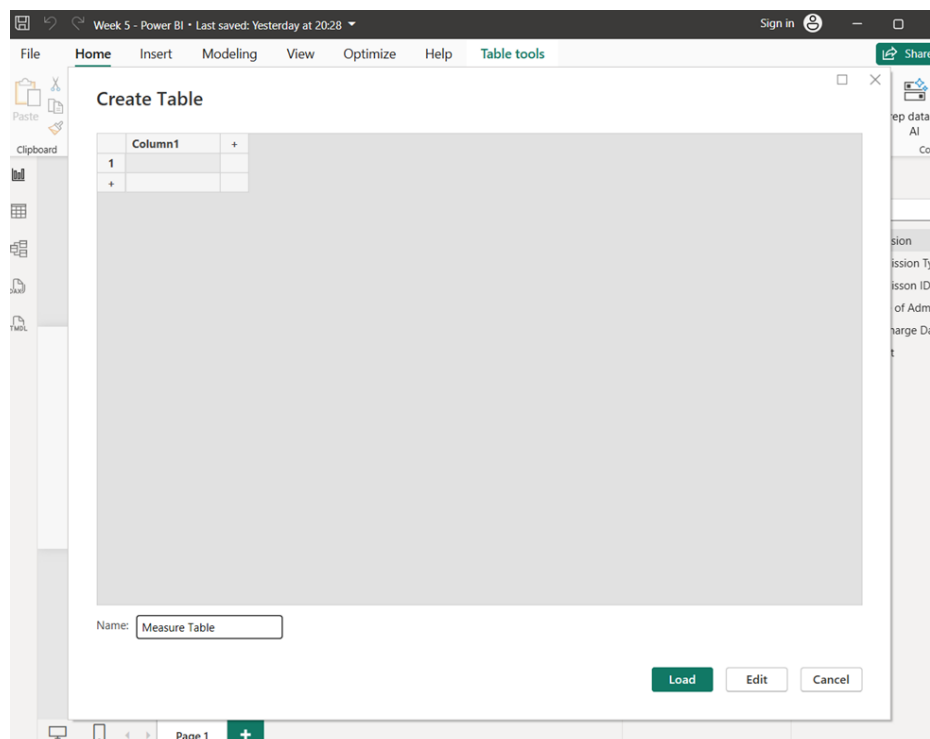
2. Objectives

- Write DAX measures for Total Patients, Total Billing Amount, Average Billing Amount, Average Length of Stay and Readmission Rate.
- Build top KPI summary cards for the main numbers.
- Show patient demographics using a matrix visual and a donut chart.
- Show medical conditions and admission volumes using a stacked bar chart.
- Show admission trends over time using a line chart, and billing by insurance provider using a column chart.
- Add slicers for Insurance Provider, Admission Type and Date of Admission for dynamic cross-filtering.

3. Steps

Step 1: Create a Measure Table

Before writing the measures, I made a separate blank table called Measure Table using Enter Data, so that all my DAX measures stay in one place instead of mixing with the Fact_Billing columns.



Step 2: Write the DAX measures

I wrote the following measures in the Measure Table:

```
Total Patients = COUNTROWS(Fact_Billing)

Total Billing Amount = SUM(Fact_Billing[Billing Amount])

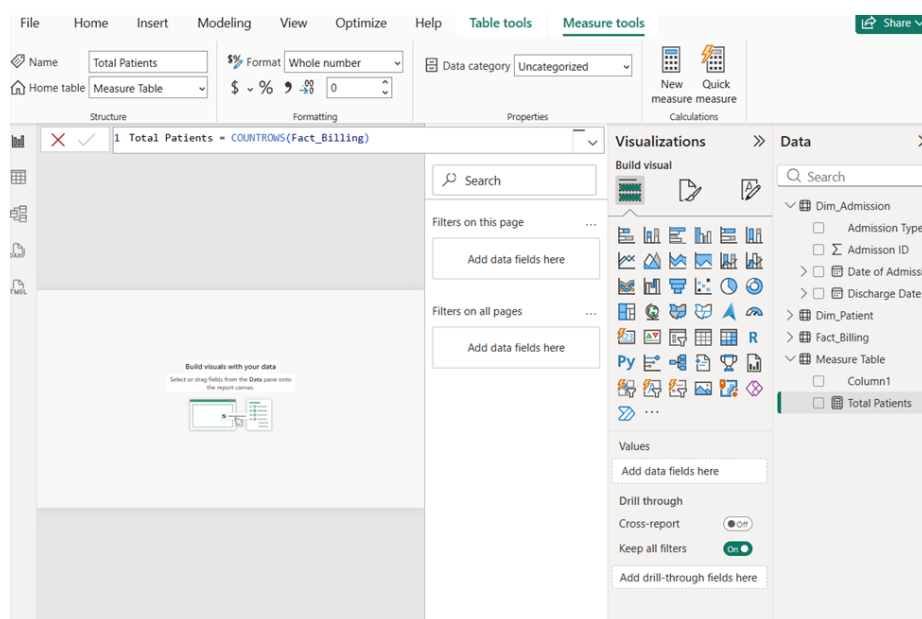
Avg Billing Amount = AVERAGE(Fact_Billing[Billing Amount])

Avg Length of Stay =
    AVERAGEX(Fact_Billing,
        DATEDIFF(Fact_Billing[Date of Admission], Fact_Billing[Discharge Date],
            DAY))
```

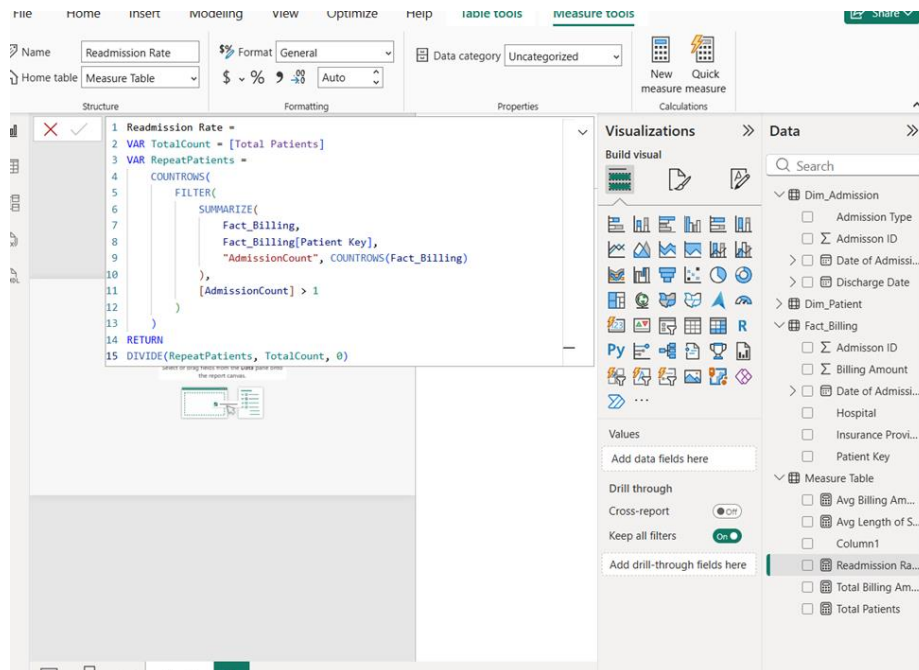
For the readmission rate, I checked how many patients appear more than once in Fact_Billing using their Patient Key:

```
Readmission Rate =
VAR TotalCount = [Total Patients]
VAR RepeatPatients =
    COUNTROWS(FILTER(
        SUMMARIZE(Fact_Billing, Fact_Billing[Patient Key],
            "AdmissionCount", COUNTROWS(Fact_Billing)),
        [AdmissionCount] > 1))
RETURN DIVIDE(RepeatPatients, TotalCount, 0)
```

The Total Patients measure came out correct in the Data pane, counting every row in Fact_Billing:

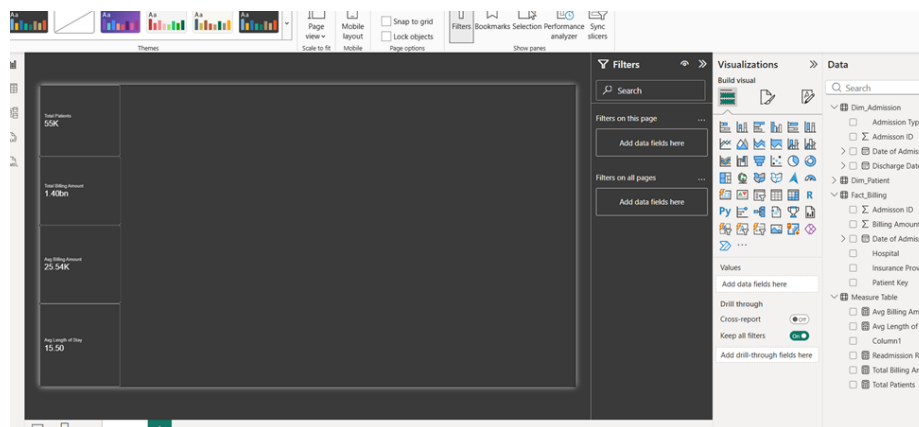


The Readmission Rate measure also worked and showed up under the Measure Table:



Step 3: Build the top KPI cards

I placed four Card visuals at the top of the page for Total Patients, Total Billing Amount, Avg Billing Amount and Avg Length of Stay, so the main numbers are visible right away.



Step 4: Demographics and patient distribution

I added a Matrix visual with Age Group and Gender in the rows, Medical Condition in the columns, and Total Patients plus Avg Billing Amount as the values. Next to it I placed a Donut Chart with Age Group in the legend and Total Patients as the value, so I can see the split of patients by age at a glance.

Step 5: Medical conditions and admission volumes

I made a Stacked Bar Chart with Medical Condition on the Y-axis and Total Patients split by Admission Type on the X-axis, so I can compare how many Elective, Emergency and Urgent admissions happen for each condition.

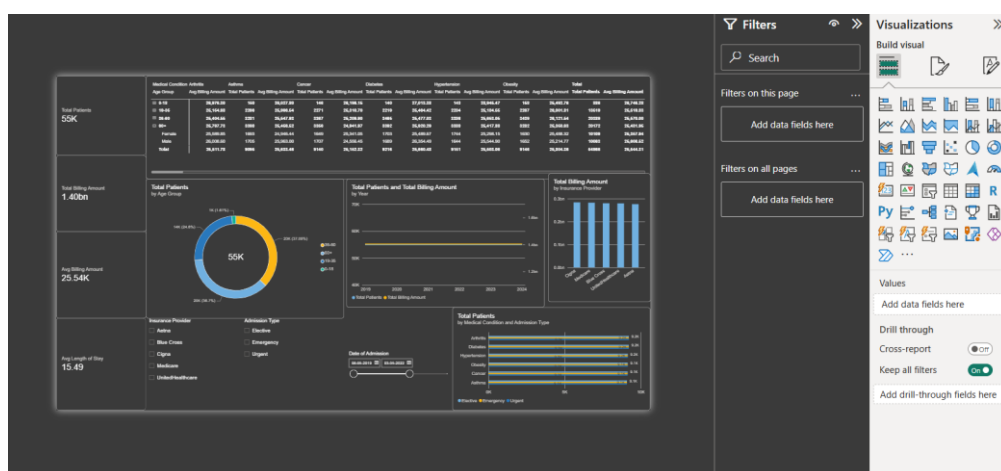
Step 6: Admission trends and hospital analytics

I added a Line Chart with Date of Admission (Year, Month) on the X-axis and Total Patients plus Total Billing Amount on the Y-axis, to see how admissions and billing change over time. I also added a Clustered Column Chart with Insurance Provider on the X-axis and Total Billing Amount on the Y-axis, to compare billing across providers.

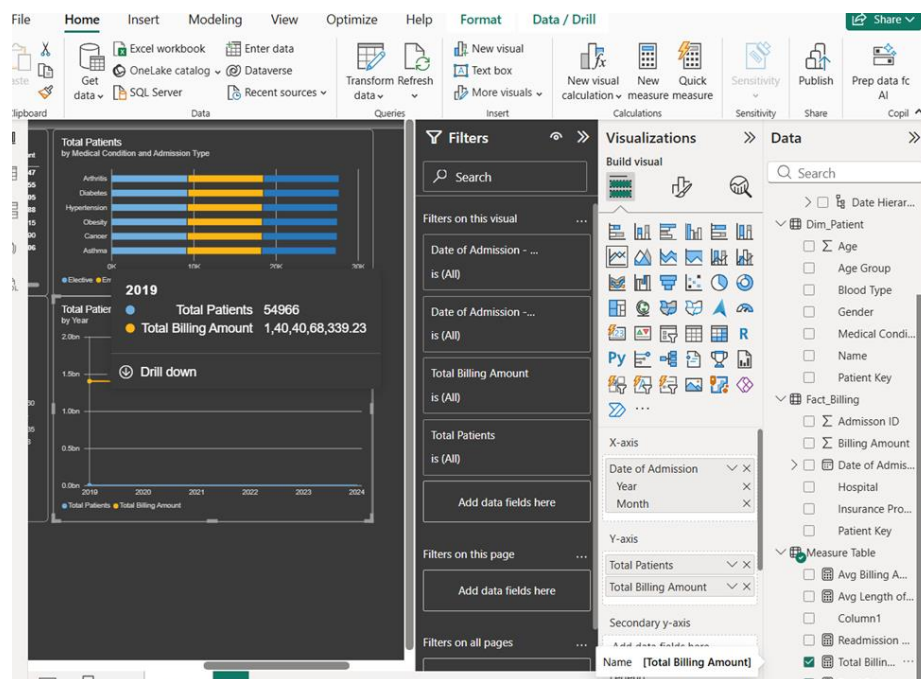
Step 7: Slicers and cross-filtering

Finally, I added three slicers: Insurance Provider (tile), Admission Type (tile) and a Date Range slicer for Date of Admission. These slicers cross-filter every visual on the page, so selecting one insurance provider updates the cards, the matrix and all the charts together.

Here is the finished dashboard:



Hovering over the line chart also shows a tooltip with the exact numbers, and lets me drill down from year to month:



4. Observations

- The total patient count is around 55K, with an average billing amount of about 25.5K and an average length of stay of about 15 days.
- Patients are fairly evenly spread across the 19-35, 36-60 and 60+ age groups, with a much smaller 0-18 group.
- Billing amount does not change much between different medical conditions, since the average billing per condition is close to the overall average.
- Total billing amount by insurance provider looks almost equal across all five providers, so no single provider stands out.
- The slicers make it easy to filter the whole dashboard by a specific insurance provider, admission type or date range without touching each chart separately.

5. Conclusion

This week I learned how to write DAX measures for KPIs like total patients, billing amount, length of stay and readmission rate, and keep them organized in a separate Measure Table. I also learned how to plan and build a dashboard layout using KPI cards, a matrix, a donut chart, bar and line charts, and how slicers make the whole page interactive. Overall, this task helped me understand how a clean data model from Week 5 can be turned into a useful, easy-to-read healthcare dashboard.